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Magnetostratigraphic dating of the proposed Rhaetian GSSP at Steinbergkogel (Upper Triassic, Austria): Implications for the Late Triassic time scale

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ABSTRACT

The Global Stratotype Section and Point (GSSP) for the Rhaetian Stage has recently been proposed at Steinbergkogel in Austria. We re-sampled the Steinbergkogel sections (STK-A and STK-B + C) in high-resolution to establish a robust magnetostratigraphy that allows global correlation. The palaeomagnetic signal at Steinbergkogel is composed of three components, which can be separated by thermal demagnetization. The highest temperature component, revealed between 280/300 and maximum 600 °C is of dual polarity and is interpreted as primary. Rock magnetic experiments showed that the signal is carried by magnetite. Our results allow correlation between the two individual Steinbergkogel outcrops. Subsequently, we correlate the two key biostratigraphic horizons for the base of the Rhaetian, the FO of *M. hernsteini* and the FAD of *M. post-hernsteini* to other sections of the Tethys domain. The correlation to the astronomically dated continental successions of the Newark basin indicates that these positions for the base of the Rhaetian are most likely determined in chrons E16n and E16r, respectively. This correlation is confirmed by cyclostratigraphic control on the marine Pizzo Mondello (Italy) section, where a combination of long period Milankovitch cycles (~175-Myr) and short-eccentricity cycles (~100-kyr) provide additional correlation constraints, respectively supporting a long duration of the Rhaetian. Our study implies that the Norian and Rhaetian Stages have durations of ~17 and ~9 Myr.

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1. Introduction

The Late Triassic period is characterized by increasing environmental stress that eventually culminated in a mass-extinction in the latest Rhaetian. This end-Triassic extinction is one of the five major mass-extinctions that have taken place on Earth and is generally considered to be related to massive volcanism in the Central Atlantic Magmatic Province (CAMP) (Deenen et al., 2010; Hesselbo et al., 2002; Marzoli et al., 2004) corresponding to environmental changes in marine and continental ecosystems. The extinction paved the way for the dinosaurs to become the dominant species on Earth. Understanding the timing and pace of Late Triassic extinctions is hampered by the poor quality of the available age constraints. Radiometric ages are scarce and magnetostratigraphic correlations are generally debated. Changes in the Geological Time Scale (GTS) give rise to additional misunderstandings. Two different definitions have been used for the base of the Rhaetian (middle Sevatian substage, or top Sevatian). Furthermore, marine–continental correlations are generally considered controversial. In particular, the position of the Norian–Rhaetian boundary in continental successions is highly debated, which results in contrasting estimates of the duration of the Rhaetian stage of 1 and almost 7 Myr (see for instance Hounslow and Muttoni, 2010). In order to establish a standard usage for this boundary and other stage boundaries, the International Commission on Stratigraphy (ICS) has established Global Stratotype Sections and Points (GSSPs) in boundary reference sections, and its Triassic subcommission has been working on GSSPs for the bases of the Norian and Rhaetian stages. The GSSP defining the base of the Hettangian (base Jurassic) was formally accepted by ICS and was awaiting official ratification by the International Union for Geological Sciences (IUGS) at the time of publication of our study. The GSSP for the base of the preceding Rhaetian stage has been proposed at a nearby Austrian section at Steinbergkogel (Krystyn et al., 2007a, b). A magnetostratigraphy was published for subsections A and B + C (Krystyn, 2008b; Krystyn et al., 2007a,b).

We re-sampled these two sections at Steinbergkogel in high-resolution to establish a robust magnetostratigraphic record that would allow internally consistent correlations and correlations to other magneto-bio-stratigraphically dated sections in the Tethys domain; Silická Brezová (Channell et al., 2003), Pizzo Mondello (Muttoni et al., 2004), Kavur-Tepe (Gallet et al., 1993), Scheiblkogel (Gallet et al., 1996) and Costa Imagna–Brumano–Italcementi Quarry composite section (Muttoni et al., 2010). The correlation to the continental Newark Basin successions – which are astronomically calibrated to the radiometrically dated CAMP basalts (201.27 ± 0.06 Ma; Schoene et al., 2006, 2010) – allows calculation of the age and duration of the Norian and Rhaetian stages. We furthermore propose an integrated stratigraphy approach to

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solving the Late Triassic time scale and use cyclostratigraphic data from Pizzo-Mondello to validate our results. Finally, we discuss the consequences of our magnetostratigraphic correlations for the Late Triassic time scale.

2. The Steinbergkogel section: geological setting and background data

The Steinbergkogel section is located near Hallstatt, in the Salzkammergut, Austria (47°33′53″N, 13°37′33″E) and is exposed in the cliffs of an abandoned quarry (Krystyn et al., 2007a,b) (Fig. 1). The section is characterized by Hallstatt-type limestones, which were deposited at low palaeolatitude in the Late Triassic Tethys Ocean. The limestones are occasionally intercalated with marls, which grade upwards into the grey marls of the Rhaetian Zlambach Formation (Krystyn et al., 2007a, b). The nodular limestones and nodular appearance of the marly layers indicate a low sedimentation rate and compaction after deposition. The transition to the Zlambach Formation most likely coincides with an increase in sedimentation rate (Krystyn et al., 2007a).

The Steinbergkogel quarry consists of two subsections: the ~3.5 m STK-A section on the eastern side of the outcrop and the ~11 m STK-B + C section to the west (Fig. 1b, c). The bedding orientation at Steinbergkogel shows a steep dip towards North (strike/dip; 257/63 N). The Hallstatt-facies interval is characterized by thinning of the layers towards the west and additional layers are exposed along the eastern side (see Fig. 1c). While logging section STK-A and STK-B + C, we followed the numbering scheme for individual layers of Krystyn et al. (2007b), which was slightly modified by Krystyn (2008b). However, based on field observation, our STK-A is slightly extended upward including layers 123 to 125, but does not include layers 103 to 107 at the base. These layers are in the basal part of section STK-B + C. Earlier established magnetostratigraphy of both subsections yielded a polarity pattern consisting of five normal and four reverse polarity intervals in section STK-B + C and four normal and three reverse polarity intervals in section STK-A (Krystyn et al., 2007b, updated in Krystyn, 2008b). The palaeomagnetic correlation between the two sections was not entirely straightforward (Krystyn, 2008b) as one normal and reverse chron was considered missing in section STK-B + C, and palaeomagnetic results from layers 120 to 122 in section STK-A were uncertain due to a combination of declinations of 0–90° and negative inclinations (Krystyn et al., 2007a). In addition, the positions of biostratigraphic events relative to polarity intervals in the two subsections were not fully consistent (Krystyn et al., 2007b). Nevertheless, these palaeomagnetic results indicated the potential for establishing an improved Rhaetian magnetostratigraphy at Steinbergkogel, if sampled at high-resolution.

The Rhaetian GSSP is proposed in the STK-A section (Krystyn et al., 2007a,b) with three biostratigraphic options that have been put forward. In option 1, the Norian–Rhaetian boundary is defined by the First Occurrence (FO) of *Misikella hernsteini*, a conodont species that is easy recognizable and widely present in low palaeolatitude regions of the Tethys Ocean. In the STK-B + C section, this level is also marked by the FO of *Oncodella paucidentata* and *Epigondolella mosheri* morphotype A of Orchard et al. (2007) (Krystyn et al., 2007b). In option 2, the Norian–Rhaetian boundary is defined by the First Appearance Datum (FAD) of *Misikella posthernsteini*, the phylogenetic successor of *M. hernsteini*. In both options, the Norian–Rhaetian boundary is closely associated with the Sevatian 1–Sevatian 2 boundary (Krystyn et al.,

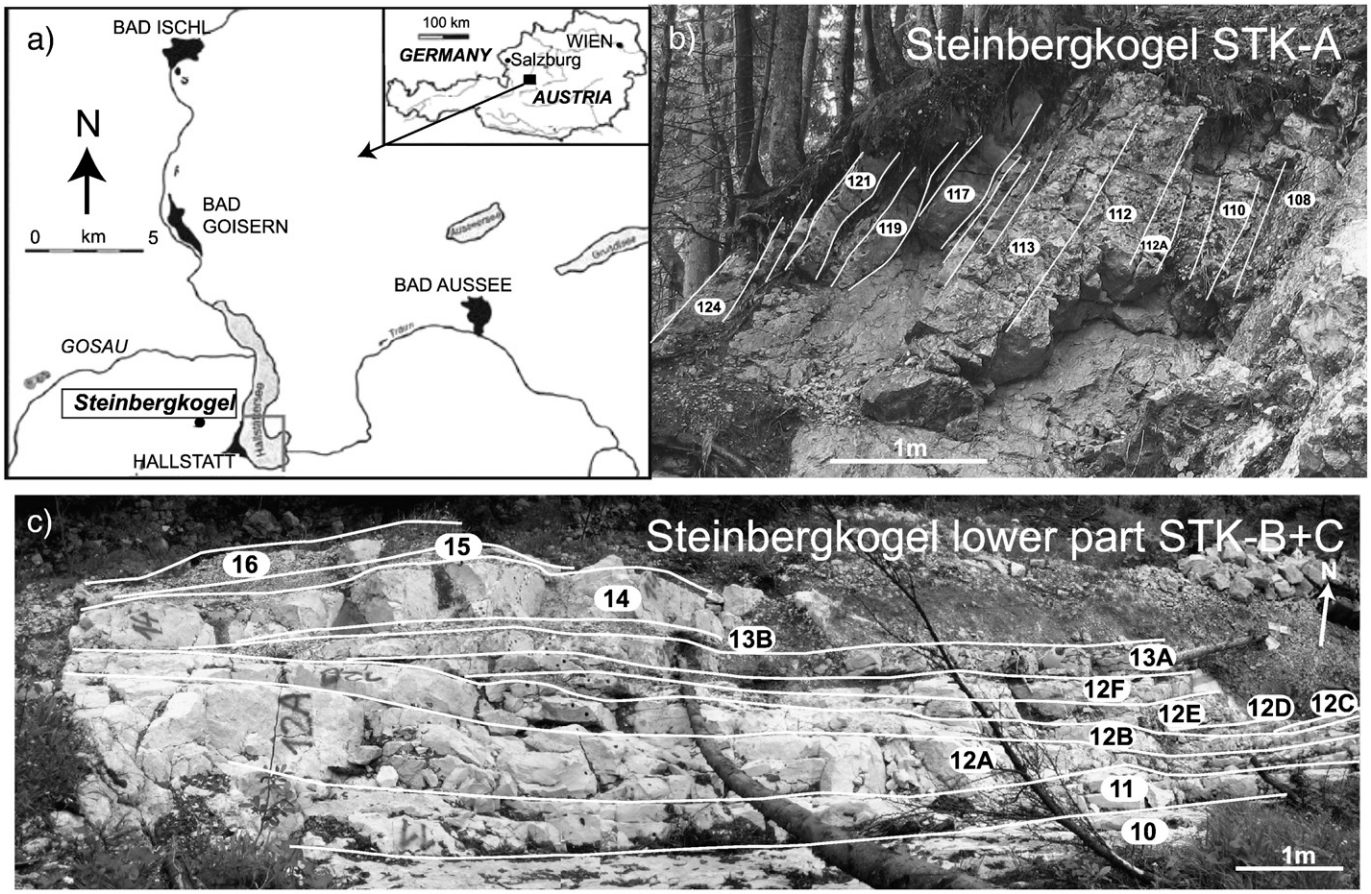


Fig. 1. a) Overview of the location of the Steinbergkogel section in Austria (map modified after Krystyn et al. (2007b). b) Photograph of the STK-A subsection of Steinbergkogel, c) photograph of the lower part of the STK-B + C section of Steinbergkogel. Numbers in the photographs correspond to the numbers of layers indicated in Fig. 4 following Krystyn et al., 2007b.

2007b). The third option to define the Norian–Rhaetian boundary is related to the First Appearance (FA) of the ammoid *Vandaites*, a very prominent event that coincides with other well-defined first occurrences and disappearances (Krystyn et al., 2007b). This event, however, cannot be correlated to a reversal boundary in the magnetostratigraphic succession of Steinbergkogel, therefore lessening its correlation potential. The Norian–Rhaetian Task Force has proposed the FAD of *M. posthernsteini* as marker for the base of the Rhaetian (Krystyn, 2008a; Krystyn et al., 2007a).

3. Palaeomagnetic results

3.1. Rock magnetism

Rock magnetic analyses were carried out on selected samples to understand the magnetic properties of the Steinbergkogel rocks before applying them to demagnetisation methods. First, Isothermal Remanent Magnetization (IRM) acquisition was performed by applying 56 peak fields up to 700 mT. The imparted IRM was subsequently measured on a robotized horizontal 2G Enterprise DC SQUID magnetometer (noise level $1–2\times10^{-12}$ Am^2). Each sample was alternating field (AF) demagnetized at 300 mT prior to IRM acquisition so that they were in the AF demagnetized starting state (Heslop et al., 2004). All data were subjected to component analysis of cumulative log-Gaussian (CGL) curves (Kruiver et al., 2001), where every IRM curve was decomposed into a number of CGL curves. Two components were distinguished, which can be individually characterized by their saturation IRM (SIRM), remanent acquisition coercive force ($B_{1/2}$), and dispersion parameter (DP).

Component 1 is fitted for all samples in the medium field interval, with a $B_{1/2}$ value of ~32 mT and a DP of ~0.29 $\log_{10}$ mT indicating a mixed grain size distribution (Fig. 2). The $B_{1/2}$ values are typical of

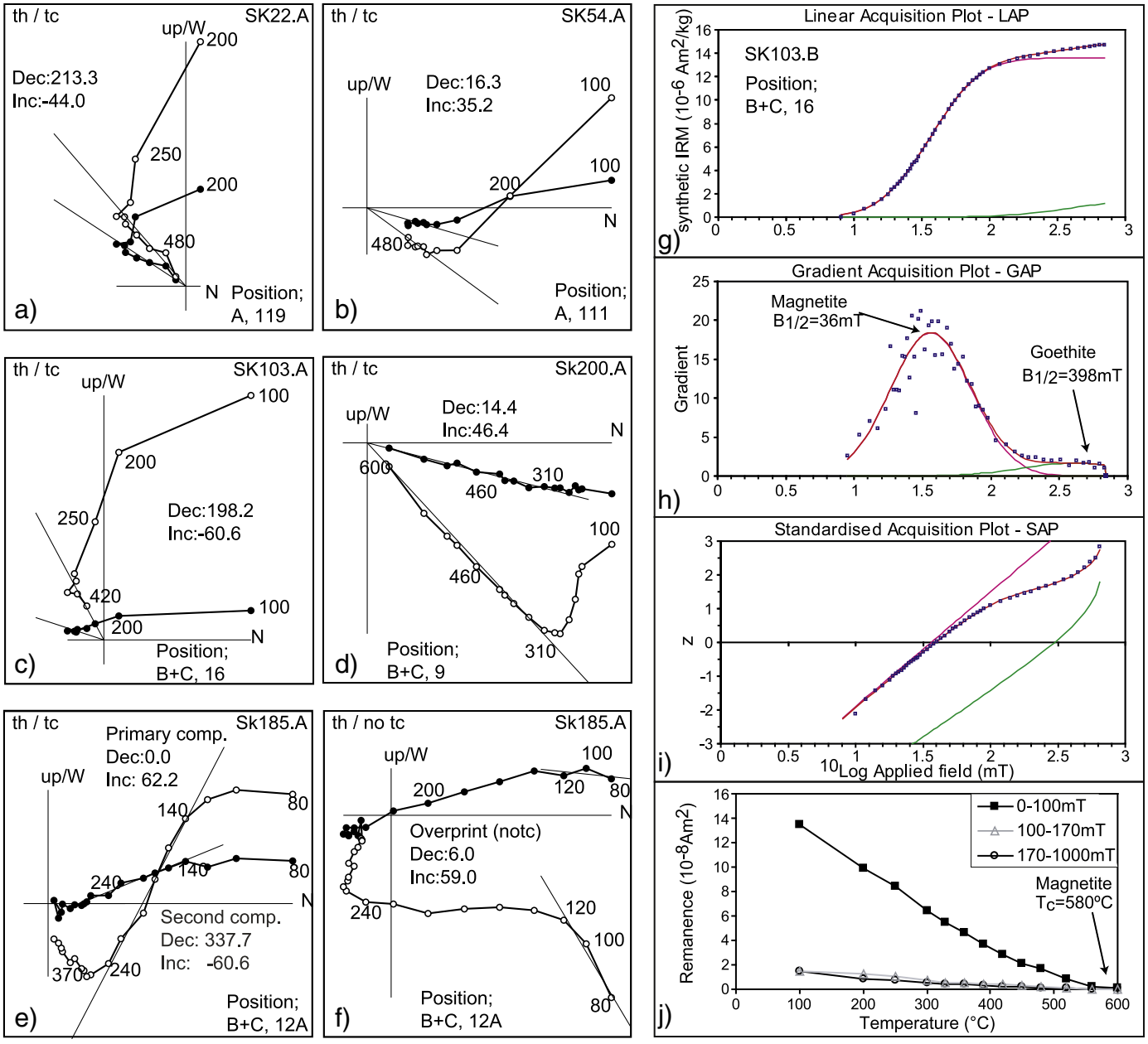


Fig. 2. a–f) Representative Zijderveld diagrams of thermal demagnetization (th). Stratigraphic positions are given in the diagrams. The room temperature step (20 °C) is omitted from these representations, because it shows only a randomly oriented viscoud component (see text) a–b) Diagrams are from section STK-A; c–f) diagrams are from STK-B + C; the numbers next to the graph denote temperature steps in °C; g–j) IRM acquisition and thermal demagnetization of IRM of representative sample SK103.B. See text for discussion on magnetite and goethite components. (no)tc = tectonically (un)corrected. Tc = Curie temperature.

magnetite and the mixed grain size distribution is indicative of a detrital origin. This magnetite component contributes about 86% of the total IRM and is therefore the main magnetic carrier. Component 2 is fitted in the high-field interval and has $B_{1/2}$ values 390 and 950 mT and a large DP of ~0.34 $\log_{10}$ mT. Its presence is also indicated by the high-field 'tail' observed in the standardized acquisition plots (SAP). This component is responsible for the non-saturation of IRM and could be carried by either haematite or goethite. The observed NRM decrease below 600 °C precludes the presence of haematite. Goethite is likely present as a product of weathering processes (Dunlop and Özdemir, 1997; France and Oldfield, 2000; Kruiver et al., 2001; Maher and Hallam, 2005). Its contribution to the total IRM is only minor with on average 12%. The $B_{1/2}$ values are only indicative because the applied maximum field of 700 mT is not sufficient for goethite saturation.

Results of the CGL analysis are further used to select suitable peak fields for three-axis thermal demagnetization of the IRM (Lowrie, 1990). Fields of 1000, 170 and 100 mT were subsequently imparted, followed by stepwise thermally demagnetization up to 600 °C. For all samples, the low coercivity fraction (0–100 mT) is a constant and main contributor to the IRM (Fig. 2). This coercivity range is indicative of magnetite, maghemite, titanomagnetite, greigite and/or pyrrhotite. Based on CGL analysis and thermomagnetic data we can rule out the presence of maghemite, titanomagnetite, greigite and pyrrhotite. The medium (100–170 mT) and hard (170–1000 mT) fractions contribute only a small percentage to the total remanence. The remanence of the low coercivity fraction (0–100 mT) linearly decays up to 600 °C with a small remanence increase at ~480 °C. The maximum unblocking temperature of this fraction just below 600 °C indicates the presence of relatively pure magnetite. The relatively small remanence increase between 450 and 500 °C in the low coercivity fraction may indicate pyrite oxidation (Passier et al., 2001). Hence, we conclude that magnetite is the magnetic remanence carrier throughout the entire Steinbergkogel section.

3.2. NRM demagnetization

The Steinbergkogel section was re-sampled with a resolution of one oriented sample about every 5 cm throughout the STK-A and STK-B + C sections. One specimen per sample level was thermally demagnetized in a magnetically shielded furnace using small temperature increments of 20 to 30 °C up to 600 °C. The natural remanent magnetization (NRM) was measured on a 2 G Enterprise horizontal cryogenic magnetometer equipped with DC SQUIDS (noise level 3×10^{-12} Am²). The directions of the NRM components were calculated by principal component analysis (Kirschvink, 1980; Zijderveld, 1967). The 63° tilt will evidently help to distinguish primary (pre-tilt) from secondary (post-tilt) components and to recognize present-day field overprints.

Initial NRM intensities (at 20 °C) range between 0.0038 and 0.2882 mA/m. Thermal demagnetization with linear decay towards the origin are considered reliable resulting in approximately 55% of the demagnetization diagrams of good quality (Fig. 2). In general, the quality of the thermal demagnetization is better for the STK-B + C than for STK-A section. During thermal demagnetization, a randomly oriented viscous component is removed between room temperatures (20 °C) and 80 °C (Fig. 3). Further heating reveals the removal of a low-temperature component between 80 and 160/180 °C. Without tectonic tilt correction, the directions of this component cluster close to the present-day field direction of Steinbergkogel, with mean directions of Dec = 0.3° and Inc = 49.0° (expected direction at location Dec = 2° and Inc = 63°) (Fig. 3). We interpret these directions as secondary post-tilt overprints. A medium-temperature component is removed between 160/180° and 280/300 °C. Before tectonic tilt correction, the mean direction of this component is Dec = 339.6° and Inc = 0.4°. After tilt correction, these values change to Dec = 332.3° and Inc = −61.3° and plot in a well-defined circle with little scatter on the equal area projection (Fig. 3). This component is interpreted as secondary overprint, although the unusual combination of a northward declination (332.3°) and large negative inclination (−61.3°) does not conform to normal or reverse overprint directions. A high-temperature component is generally removed from 280/300 °C to maximum temperatures of 600 °C. After tectonic tilt correction, this component progressively decays towards the origin and shows dual polarity. The normal polarity samples have a mean direction of Dec = 13.9° and Inc = 40.9°, the reverse polarity samples have Dec = 223.9° and Inc = −57.8°. Based on the dual polarity pattern and the significant difference to the present-day field directions, we interpret this Characteristic Remanent Magnetisation (ChRM) as a primary component, even though the normal and reverse directions are not antipodal. (The palaeomagnetic data is made available in the online supplementary data file.)

The complex NRM behaviour indicates that three components are present with overlapping blocking temperatures. This caused mixed directions of the medium- and high-temperature components, which proved to be difficult to isolate. Rock magnetic experiments indicate that all three components are carried by the same magnetic remanence carrier, magnetite.

3.3. Magnetostratigraphy

The directions of the high-temperature ChRM component are plotted in stratigraphic order, resulting in magnetic polarity patterns for both subsections (Fig. 4). The polarity column of STK-A section is characterized by four normal and four reverse polarity intervals. Unfortunately, it also contains four intervals of uncertain polarity and polarity boundaries within these uncertainty intervals are therefore not well defined. The STK-B + C section reveals a long normal interval at the base (~3 m), followed by an interval dominated by reverse polarity (~5 m), which contains three to four short normal intervals. From ~8 m upward, an uncertain polarity interval is present, where directions and reversal boundaries are unfortunately not well delineated. The uppermost ~2 m – part of the Zlambach Formation – are entirely characterized by normal polarity. Samples that displayed uncertain ChRM directions nevertheless follow the general trend of the magnetic polarities throughout the section.

Our new palaeomagnetic results replicate the main features of the polarity pattern of the previously established magnetostratigraphy (Krystyn et al., 2007b). The increased resolution indeed clarifies the previously normal polarity zones in layer 14C and 23B in section STK-B + C, which had been suggested by one sample only (Krystyn, 2008b; Krystyn et al., 2007b). Our results provide important details of a polarity subchron within the long reverse-dominated interval, such as the short normal chron in layer 16 in section STK-B + C. In the upper part of section STK-A, we define an additional reversal boundary, which indicates the short duration of the normal chron in layers 121 and 122, which is followed by reverse polarity. In general reversal boundaries in section STK-B + C are better delineated, which improves the correlation between both subsections. We can thus correlate the short normal chron (a) in layer 12 to that in layer 111 in section STK-A, (b) in layer 14 to that in layer 116–117 in section STK-A, and (c) in layer 16 to that in layer 121–122 in section STK-A.

The palaeomagnetic correlation between the two subsections, STK-A and STK-B + C, is further supported by biostratigraphic data and field observations. The FO of *M. hernsteini* in the upper part of the normal polarity interval at the base of STK-B + C and STK-A sections confirms our correlation. The FAD of *M. posthernsteini* is located within the successive reverse interval, although the exact position differs between the two sections, possibly related to bio-magnetostratigraphic sample resolution (Fig. 5). Upward correlation indicates that the STK-A section corresponds to the long interval dominated by reverse polarity, and the short normal polarity intervals correlate very well (Figs. 4 and 5).

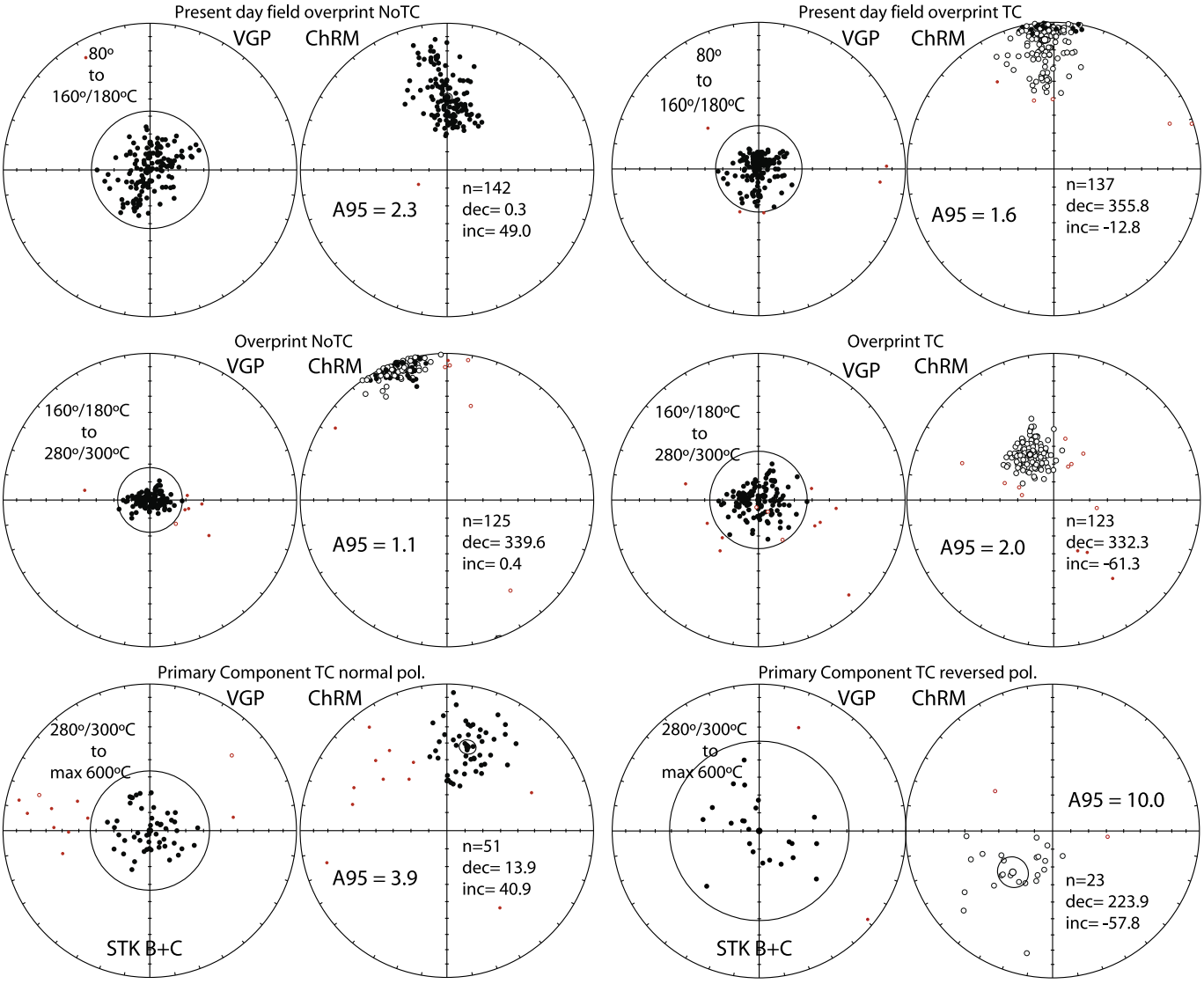


Fig. 3. Equal area projections of the three components isolated from the demagnetisation diagrams in (un)corrected – (no)tc – positions.

4. Implications for the Late Triassic time scale

4.1. Magneto-biostratigraphic correlation within the marine Tethys domain

Now that we have established a magnetostratigraphy for the Steinbergkogel sections we first investigate the magneto-biostratigraphic correlations with other Tethyan sections. The marine biostratigraphic events FO of *M. hernsteini* and/or FAD of *M. posthernsteini* – delineating the Norian–Rhaetian boundary – are considered crucial for correlation purposes. We thus first selected sections that have good biostratigraphic control and/or constitute a relatively large number of polarity reversals. Key sections for correlation are the Pizzo Mondello section on Sicily, the Silická Brezová section in Slovakia and the Kavur Tepe section in Turkey.

The ~450 m thick Pizzo Mondello section consists of deep marine cherty limestones, spanning the Late Carnian to earliest Rhaetian and has a high-resolution magnetostratigraphy (Muttoni et al., 2001, 2004). The Norian–Rhaetian boundary was tentatively placed on top of normal polarity zone PM11n, at the base of the first calcilutite bed containing the FO of *M. posthernsteini* (Gullo, 1996). The preceding FO of *M. hernsteini* is positioned within or just above the top of PM11n, but was not well-delineated. These conodont events show that PM11n correlates to the basal normal polarity zone at Steinbergkogel (Fig. 6). We consider this correlation in better agreement with the biostratigraphic markers than the correlation of the basal normal polarity zone at Steinbergkogel to PM12n at Pizzo Mondello (Hounslow and Muttoni, 2010).

The Silická Brezová section is a composite section of upper Carnian to lowermost Rhaetian marine limestones of ~125 m thickness in South Slovakia with a robust magnetostratigraphy (Channell et al., 2003). The reverse polarity zone SB-10r was assigned to the lowermost *M. hernsteini–P. andrusovi* Zone, indicating a correlation to the first reverse polarity interval containing both *M. hernsteini* and *M. posthernsteini*. Consequently, SB-10n corresponds to the basal normal polarity interval of Steinbergkogel (Fig. 6). Therefore, SB-10n at Silická Brezová correlates to PM11n at Pizzo Mondello, which is in line with the biostratigraphic markers and with the correlation proposed by Gallet et al. (2007) and Muttoni et al. (2004).

The Kavur Tepe section in Turkey consists of ~30 m condensed Hallstatt limestones of upper Norian to lowermost Rhaetian and has a detailed magnetostratigraphy (Gallet et al., 1993). The position of the FA of *M. hernsteini* is located in the top of normal Chron KT-H1 allowing correlation to the long normal at the base of the Steinbergkogel section (Fig. 6). This suggests that the short polarity zonation KT-H2−, which is originally based on one sample, represents a reverse polarity subchron. Downward correlation indicates a discrepancy in the position of the middle to Late Norian boundary

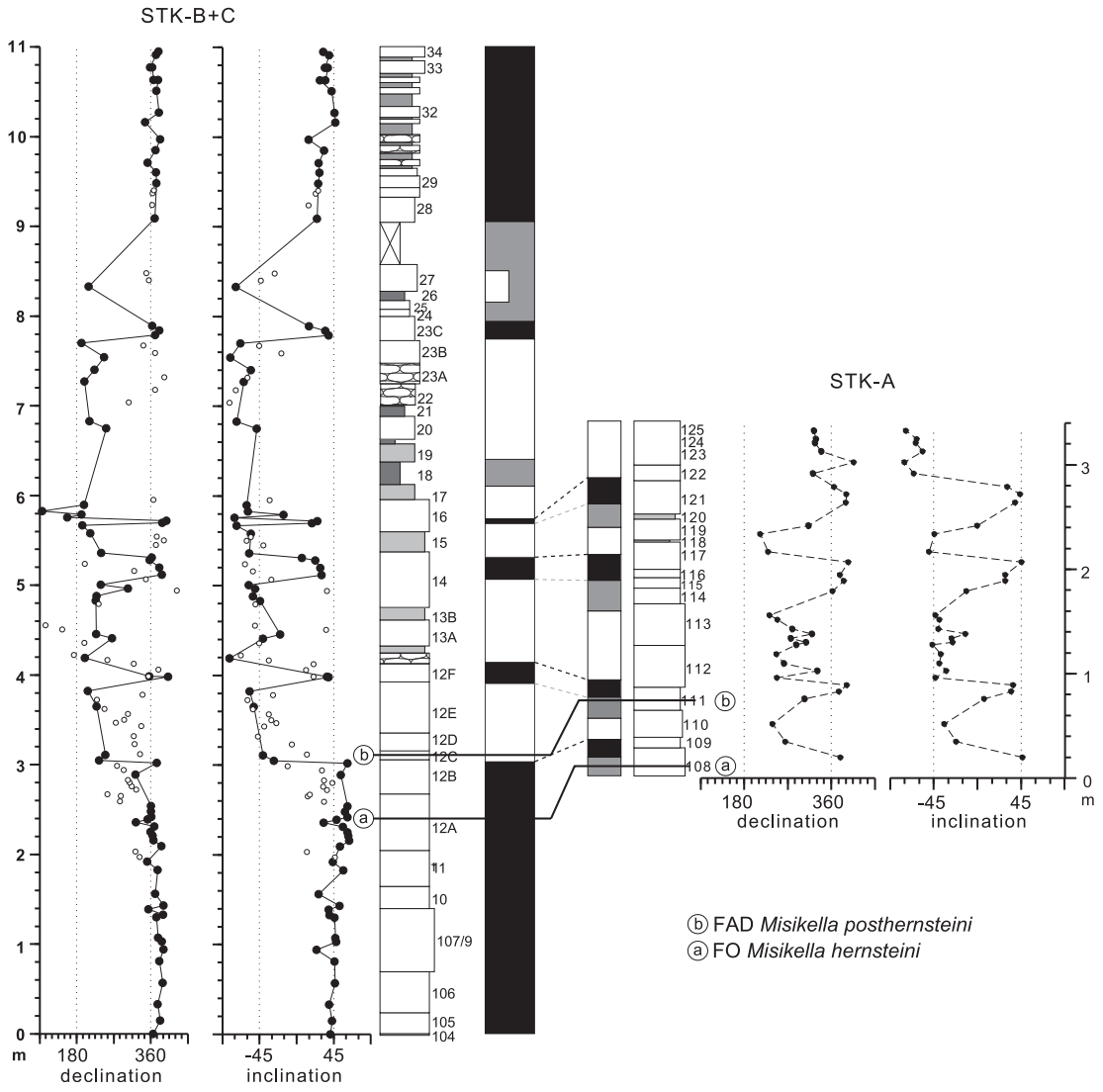


Fig. 4. Magnetostratigraphy for both subsections STK-A and STK-B + C of the Steinbergkogel section. Normal and reverse polarity is derived from declination and inclination interpretations, where closed (open) points denote reliable (unreliable) ChRM directions and in the polarity column black denotes normal polarity, white denotes reverse polarity, and grey denotes unclear polarity. Numbers next to the layers in the lithology columns follow Krystyn et al. (2007b) numbering scheme and are also indicated on the photographs in Fig. 1. The positions of the biostratigraphic events are according to Krystyn et al. (2007b).

(Alaunian–Sevatian-1). In Silická Brezová, this boundary is delineated by the disappearance of *Mockina postera* and appearance of *Mockina bidentata* within reverse Chron SB-9r which is palaeomagnetically correlated to KT-G reverse in Kavur Tepe, i.e. one reversal chron younger than proposed by Gallet et al. (1993). We follow here the magnetostratigraphic dating of Silická Brezová (Channell et al., 2003) because it seems better constrained; related to a higher sample resolution and higher sedimentation rate of the stratigraphic sequence.

We now include two additional palaeomagnetically investigated sections from the Tethyan realm. The Scheiblkogel section in Austria, in close proximity to the Steinbergkogel section, consists of ~23 m of reddish and greyish limestones with conodonts ranging from middle Norian (Alaunian 1) to Sevatian 2 (Gallet et al., 1996). Although conodont resolution is rather coarse, the first occurrence of *M. hernsteini* is well-delineated within the short reverse polarity chron 4r. We propose to correlate this reverse polarity chron 4r to the Norian–Rhaetian boundary interval in the Steinbergkogel section and thus assign the preceding normal polarity chron 4n to PM11n, SB-10n and KTH1+ of Sevatian 1 age. The downward correlation of SB-9n (Silická Brezová) to 3n (Scheibelkogel) – both have Alaunian conodont stratigraphy – confirms the stratigraphic position of the Alaunian–Sevatian boundary in Silická Brezová – as has been suggested by Channell et al. (2003) – and in Scheibelkogel. Further correlation is in good agreement with the stratigraphic position of the Lacian–Alaunian boundary in Kavur Tepe (Fig. 6).

The Oyuklu section in southwestern Turkey has been proposed to contain the complete Late Upper Norian (Sevatian 2; now early Rhaetian) to latest Rhaetian magnetic polarity sequence (Gallet et al., 2007). This section comprises in total ~30 m of condensed limestones. The position of FO *M. hernsteini* in the normal chron at the base of the section (A+), which is followed by FAD *M. posthernsteini* in the next reverse chron (B−) enables to correlate this level to the Norian–Rhaetian boundary interval in the Steinbergkogel section, as has been proposed earlier by Krystyn et al. (2007a; Fig. 6). Consequently, magnetozone A + at Oyuklu correlates to PM11n at Pizzo Mondello as suggested by Gallet et al. (2007). We then correlate PM12n to C+. This is in much better agreement with both biostratigraphic marker positions than the proposed correlation of PM12n to E+ by Gallet et al. (2007) or PM12n to A+ by Hounslow and Muttoni (2010). The upward correlation between the Steinbergkogel and Oyuklu sections is more problematic, because the Steinbergkogel section is dominated

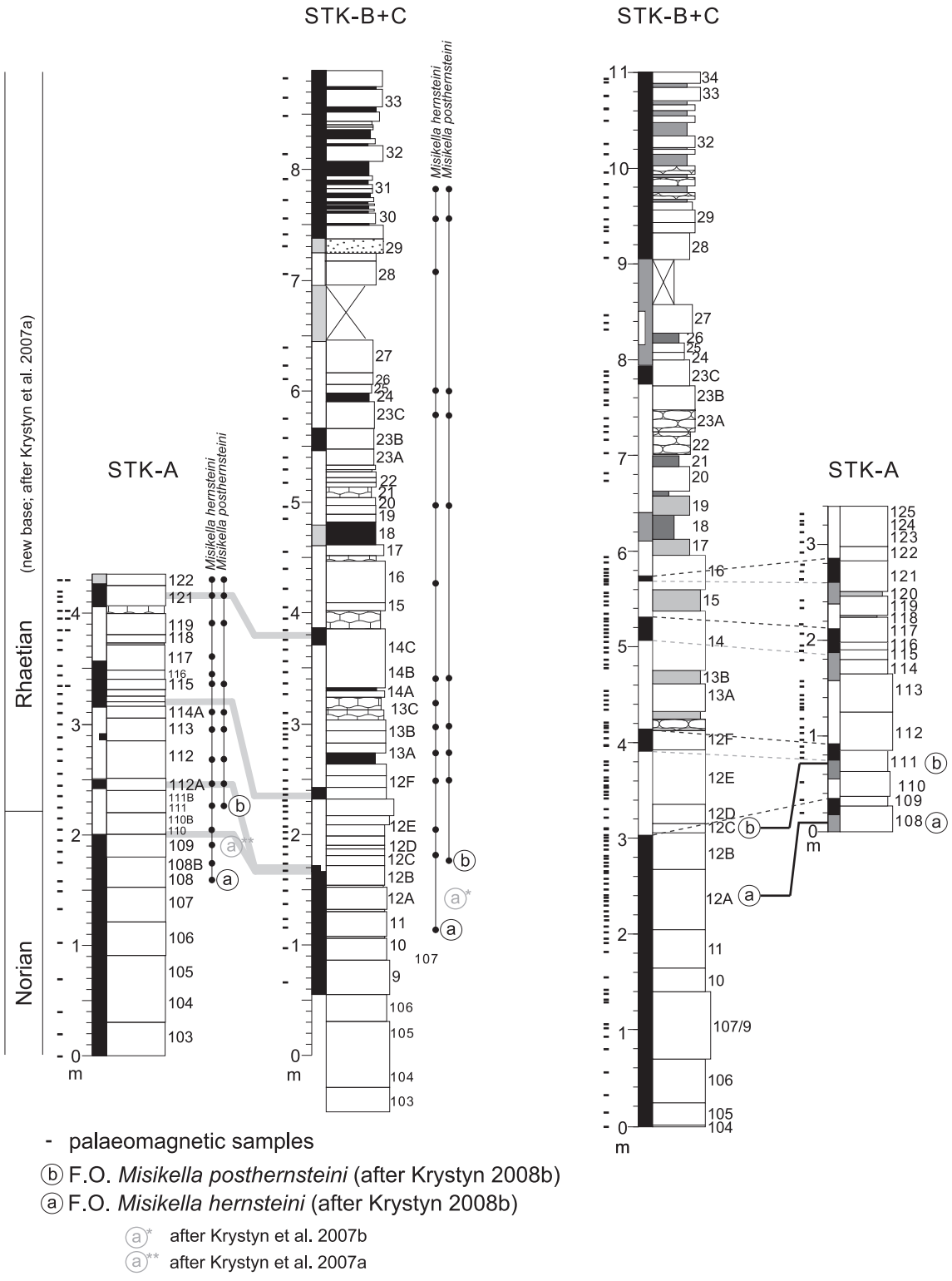


Fig. 5. Comparison between the palaeomagnetic data presented in this paper and the results shown by Krystyn et al. (2007b, 2008b). Our numbering scheme follows Krystyn et al. (2007b), and biostratigraphic positions are according to Krystyn et al. (2007b, 2008b). Note that in Krystyn et al. (2007b), both subsections contain layers 103 to 107, which are – based on our field observation – the same outcrop and therefore a repetition in the lithological log. The magnetostratigraphic correlation between STK-A and STK-B + C on the left is Krystyn's (2008b) correlation, which differs to the correlation in Krystyn et al., 2007b.

by reverse polarity, whereas the Oyuklu section is dominated by normal polarity.

The downward extension of our studied section at Steinbergkogel comprises 20 m, which fall entirely into the *E. bidentata* zone (Krystyn, 2008b) covering one single upper Norian ammonoid zone (Sagenites quinquepunctatus; Krystyn, 2008a), which is assigned to Sevatian 1. This succession is dominated by normal polarity with two short reverse chrons (Krystyn, 2008b; Fig. 6). Biostratigraphically this section can be correlated to the interval of SN9r and SB10n in Silická Brezová, the reverse and normal interval spanning the Sevatian 1 in the Scheibelkogel section (3r to 4n; Gallet et al., 1996), and KT F+ to KT H+ in Kavur Tepe (Gallet et al., 1993). However, this correlation

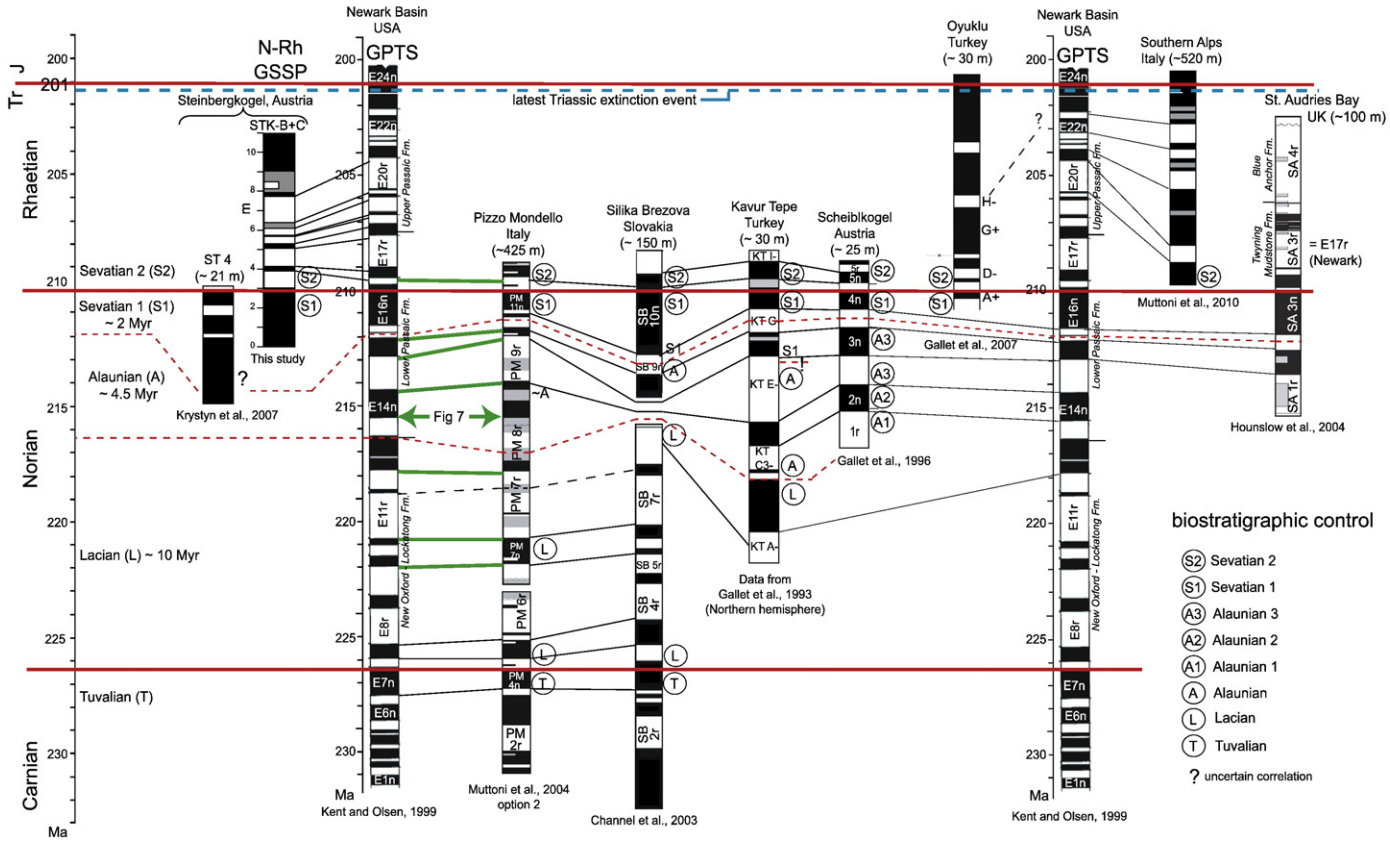


Fig. 6. Late Triassic magnetobiostratigraphic correlations. References are given in the figure. Note that we scale the Newark GPTS to the most recent CAMP date by Schoene et al. (2006) and use 201 Ma instead of previously used 202 Ma (Kent and Olsen, 1999) and 200 Ma (Channell et al., 2003).

results in an incoherent palaeomagnetic correlation between these four sections (Fig. 6).

4.2. Magnetostratigraphic correlations to the continental Newark GPTS

The magnetobiostratigraphic results of key sections of the Tethys domain are in very good internal agreement, especially for the Norian–Rhaetian boundary interval (Fig. 6). The next step is to establish the correlation of the Norian and Rhaetian stages to the orbitally controlled magnetostratigraphy of the continental Newark Basin, USA (Kent and Olsen, 1999; Kent et al., 1995), allowing calculation of their astronomical durations. The Newark Basin formed during continuous rifting between Northern America and Africa in Triassic times, due to the break-up of Pangea. The combination of palaeomagnetic research (Kent et al., 1995) and excellent control on orbital forcing of the continental deposits (Kent and Olsen, 1999; Olsen and Kent, 1996, 1999) provides a reference section for the Late Triassic GPTS.

Since the establishment of the continental GPTS in the Newark basins there has been considerable debate about the position of the Late Triassic stage boundaries, which are defined in the marine (Tethyan) realm (e.g. Gallet et al., 2000, 2003, 2007; Hounslow and Muttoni, 2010; Krystyn et al., 2007a,b; Lucas and Tanner, 2007; Muttoni et al., 2001, 2004, 2010). By now there is general consensus about the position of the Carnian–Norian boundary (Fig. 6, Krystyn et al., 2002; Muttoni et al., 2004 and Channell et al., 2003; for a more extensive review see Hounslow and Muttoni, 2010).

4.2.1. The Carnian–Norian boundary

In the Silická Brezová section, the Carnian–Norian boundary (Fig. 6) is delineated by the FAD of *Norigondolella navicula,* at the top of normal Chron SB-3n.3n (Channell et al., 2003). This biostratigraphic marker is not found at Pizzo Mondello. Instead the conodont markers FO of *Metapolygnathus echinatus* and *M. parvus* occurring within Chron PM4n have been suggested as the most suitable to define the base of the Norian (Nicora et al., 2007). At Pizzo Mondello the bounadry coincides with the $\delta^{13}C$ isotope shift in the uppermost pat of PM4n (Muttoni et al., 2004). Chron SB-3n.3n in Silická Brezová is thus correlated to Chron PM4n in Pizzo Mondello delineating the Carnian–Norian boundary in the marine realm. Channell et al. (2003) and Muttoni et al. (2004) proposed correlating PM4n and SB-3n.3n to the top of E7n (~226.6 Ma) in the Newark GPTS. This position is supported by the U–Pb zircon age of 230.91 ± 0.33 Ma from an ash bed of the upper Carnian in Italy (Furin et al., 2006) and has reached general consensus (Hounslow and Muttoni, 2010). In contrast to this generally accepted position of the Carnian–Norian boundary at Newark (Hounslow and Muttoni, 2010), Kozur and Weems (2010) suggested – based on the biostratigraphic correlation of conchostracans – placing the Carnian–Norian boundary somewhere between Smith Corner Member of the Lockatong Formation and the Warford Member of the Passaic Member in the Newark Basin succession, which is approximately at the top of E13n (~217 Ma). However, this age is in considerable disagreement with the U–Pb zircon age of 230.91 ± 0.33 (Furin et al., 2006). Moreover, as a consequence of the alternate position of the Carnian–Norian boundary in Newark (Kozur and Weems (2010), the recognition and calibration of sedimentary cycles to the orbital parameters of the Newark succession would have to be significantly revised. Additionally, based on their conchostracan biostratigraphy, more than the upper half of the Newark Passaic Formation (which has an approximate duration of 16 Myr based on the recognition of Milankovitch cycles) should be assigned to the Sevatian. This would result in a minimum duration of ~8 Myr for the Sevatian 1 and a maximum duration of ~8 Myr for the Lacian and Alaunian stages together. If this would be true, the sedimentation

rates in marine sections – where the Lacian and Alaunian stages have considerable larger thicknesses than the Sevatian (see Fig. 6) – should have changed significantly over the Alaunian–Sevatian boundary. The correlation of the Carnian–Norian boundary to the top of E13n in Newark is in disagreement with the palaeomagnetic data, because the sequence of dominantly reverse polarity in Pizzo Mondello and Silická Brezová would then correlate to the dominantly normal polarity sequence of E14n to E17n in Newark.

We consider the correlation of the Carnian–Norian boundary at the top of SB-3n.3n in Silická Brezová and at the top of PM4n in Pizzo Mondello to the top of E7n in Newark (Fig. 6) as the most robust and reliable calibration.

4.2.2. The Norian–Rhaetian boundary

The position of the Norian–Rhaetian boundary is still heavily debated (most recently summarized by Hounslow and Muttoni, 2010). Some authors (e.g. Cirilli et al., 2009; Gallet et al., 2003, 2007; Krystyn et al., 2002, 2007b) have proposed – mainly based on biostratigraphic arguments – placing the Norian–Rhaetian boundary very high in the Newark Basin sequence. This implies that either the Rhaetian is very short (ca. 1 Myr) or part of the Rhaetian is missing in the Newark Basin succession. A complicating factor in this debate is that the completeness of the continental Newark Basin succession has been frequently questioned. In particular, the biostratigraphic age assigned to the pre-basalt levels is still debated. Krystyn et al. (2002) and Gallet et al. (2003) constructed a composite GPTS considering equal duration for successive biozones, which according to them "represents a simple and convenient alternative to the large uncertainties surrounding the Upper Triassic absolute time scale and to compensate for variations in sedimentation rate among the studied sections". Krystyn et al. (2002) correlate individual sections by matching conodont ranges in order to create one multi-section composite. Short-comings of this procedure are that the selected sections are often condensed and contain stratigraphic gaps and faults (e.g. the Scheibelkogel, Kavur Tepe, Kavaalani and Bolücektasi sections; Gallet et al., 2003, 2007; Krystyn et al., 2002). Based on their correlation of the Tethyan equal-duration GPTS to the Newark GPTS Gallet et al. (2007) placed the Carnian–Norian boundary at the top of E6r. Subsequently, they suggest that E22n could lie within the uppermost Norian and that the Rhaetian is thus represented by the interval of E22r to E23n. This correlation is in disagreement with the astronomically calibrated Newark Basin timescale and implies that either the Rhaetian has a duration of only 2 Myr or that the part of the Rhaetian is partly missing in the Newark sequence (Gallet et al., 2003; Krystyn et al., 2002). A direct consequence of a hiatus in Newark is that the age of the Carnian–Norian boundary would shift downward to an older age than the cyclostratigraphic calibrated age of ~226 Ma. Gallet et al. (2003) argued that a Norian–Rhaetian boundary around E16–E18 in Newark would violate the assumption that the Middle + Upper Norian should have a much longer duration than the Rhaetian based on the highly different faunal contents. Other arguments for a potential sedimentary hiatus in the uppermost part of the Newark Basin succession come from biostratigraphic claims related to observations of the Late Triassic biotic crises in the terrestrial realm (Gallet et al., 2007 and references herein). Their points are: (1) the turnover of tetrapod and microfloral assemblages across the Triassic–Jurassic boundary are much less pronounced in Newark than in European sections (Lucas, 1994; Lucas and Tanner, 2004; Olsen et al., 1990), (2) the sporomorph taxa disappear earlier, at Sevatian–Rhaetian boundary, in Alpine sections than at the Triassic Jurassic boundary as observed in the Newark Basin (Kürschner et al., 2007), and (3) the distinct sporomorph spike previously assigned to the Triassic–Jurassic boundary was re-assigned to the Upper Norian (Sevatian), thus a Late Norian (Sevatian) age for the sediments just below the radiometrically dated CAMP lavas (e.g. Cirilli et al., 2009; Kozur and Weems, 2005, 2007; Van Veen, 1995).

A stratigraphic break just below the first basalt deposits in the Newark basin seems, however, unlikely since Deenen (2010) documented the same sequence of events (a short reverse polarity interval followed by the LO of *Patinasporites densus*) in the Argana basin (Morocco) and Fundy basin (Canada) just below the local basalt deposits as has been observed in the Newark basin. Consequent, diachronous first and last occurrences of pollen and spore taxa may be related to the climate-controlled provincialism of the parent flora and might therefore be limited for long-distance or global correlations as has been argued before (Cirilli, 2010; Krystyn et al., 2007a,b).

Others (Channell et al., 2003; Muttoni et al., 2004, 2010) have proposed a substantial Rhaetian in the Newark sequence and – mainly based on magnetostratigraphic arguments – a much longer duration for this stage (up to 8.5 Myr). In the continental Newark Basin succession, Kent et al. (1995) suggested – based on limited biostratigraphic constraints (Cornet, 1993; Fowell and Olsen, 1993) – placing the Norian–Rhaetian boundary around E18 using the old definition at the top of Sevatian 2. This correlation was later supported by the magnetostratigraphic correlation of Silická Brezová to Newark (Channell et al., 2003). Implementing the new definition of Krystyn et al. (2007a,b), the Norian–Rhaetian boundary in Silická Brezová falls just above SB-10n and therefore just above E16n in Newark.

Our preferred correlation between the Steinbergkogel section and the Newark Basin GPTS is also mainly based on their magnetostratigraphic match. Independent from previously suggested correlations, we note that the dominantly reverse polarity interval at Steinbergkogel correlates well to the interval between E16r and E20r in the Newark GPTS. This magnetostratigraphic correlation is in line with the proposed correlations of the biostratigraphically delineated Norian–Rhaetian boundary intervals in Pizzo Mondello (Muttoni et al., 2004; correlation option 2) and Silická Brezová (Channell et al., 2003) (Fig. 6). The magnetobiostratigraphic correlation of the downward extended ST4 section at Steinbergkogel to the other Tethyan sections (biostratigraphically to Sevatian 1 in SB9r to SB10n, KT F+ to KT H+, 3r to 4n) implies an anomalously expanded ST4 section with respect to our studied STK section and this succession would be correlated to E16n of the Newark GPTS (Fig. 6).

Our Steinbergkogel results – in agreement with the other Tethyan sections – delineate the Norian–Rhaetian boundary at the top of normal Chron E16n in the Newark GPTS (Fig. 6). This is the same correlation as previously proposed by Kent et al. (1995), Channell et al. (2003) and Muttoni et al. (2004) and suggests a ~6 Myr time interval for a stratigraphic sequence of ~5 m in the Steinbergkogel section. Consequently, the Norian–Rhaetian boundary has an age of ~209.8 Ma using the FO of *M. hernsteini*. The definite position of the Norian–Rhaetian boundary will evidently depend on the decision of the I.U.G.S Subcommision on Triassic stratigraphy (http://paleo.cortland.edu/sts/).

In conclusion, we deduce a long duration of the Rhaetian (from E16 to E23 in Newark) because it provides the most robust and straightforward magnetobiostratigraphic correlation between the continental Newark GPTS and the marine Tethyan realm (Fig. 6). An age of 203.5 Ma for the Norian–Rhaetian boundary (suggested by Gallet et al., 2003, 2007; Krystyn et al., 2002) is not supported. Recent palaeomagnetic results from a at least 490 m thick Rhaetian shallow marine carbonate succession of the Costa Imagna–Brumano–Italcementi Quarry composite section in the Southern Alps (Italy; Muttoni et al., 2010) corroborates the notion of a long Rhaetian and their results are in agreement with our age determination.

4.2.3. The Late Triassic substages

Upward from the Carnian–Norian boundary, the dominantly reverse polarity during the marine Lacian at Pizzo Mondello and Silická Brezová correlates very well to the dominantly reverse polarity interval E7 to E13 in the Newark GPTS. This correlation places the Lacian–Alaunian boundary (Kavur Tepe; Gallet et al., 1993), at the top of normal Chron E13n in the Newark GPTS (~216.4 Ma). The Alaunian

and Sevatian 1 magnetozones in the Tethyan sections can be correlated straightforwardly to the interval E13n to E16n in the Newark GPTS, which is characterized by normal polarity zone having longer duration than in the preceding interval E7n to E13n. The Alaunian–Sevatian 1 boundary (Silická Brezová; Channell et al., 2003) can consequently be placed within Chron E15r of the Newark GPTS (211.8 Ma). (Note that we assume no major sedimentary gaps in the marine Tethyan sections and that no magnetic polarity zones are missing along potential faults; a probable exception might the breccia interval in the Pizzo Mondello section (Muttoni et al., 2004)).

The magnetostratigraphic correlation to the Newark GPTS also provides durations for the Late Triassic (sub)stages. The early Norian (~10 Myr) is by far the longest; twice the length of the middle Norian (~4.5 Myr), and ~five times longer than the Late Norian substage (~2 Myr). This is in good agreement with the observation that the lower Norian is considerably thicker than the middle Norian in all studied Norian pelagic sequences (Gallet et al., 2007). The location of the Lacian–Alaunian boundary on the top of chron E13n is in agreement with the observation that polarity chron E12r (Newark) is still of Lacian age (Fig. 6 and Channell et al., 2003). The duration of the entire Norian therefore becomes ~17 Myr. The adopted age model of the Newark GPTS results in a duration of ~9 Myr for the Rhaetian stage and ages of ~226.6 Ma for the Carnian–Norian boundary and ~209.8 Ma for the Norian–Rhaetian boundary.

4.3. Consequences for the St. Audrie's Bay section, UK

The mostly non-marine successions of the St. Audrie's Bay section in the United Kingdom contain about 100 m of mainly red dolomitic mudstones of continental deposits (Hounslow et al., 2004). The section has limited biostratigraphic control but a relatively straightforward palaeomagnetic correlation to the interval between E14r and E20r of the Newark Basin GPTS (Fig. 6 and Hounslow et al., 2004). This correlation has been confirmed by cyclostratigraphic control (Kemp and Coe, 2007). The palaeomagnetic pattern of the St. Audrie's Bay section was correlated to the interval comprising NM3 to NM7 of the 'Tethyan equal-biozone composite' of Gallet et al. (2000) and Krystyn et al. (2002). As a result, the entire non-marine part of the St. Audrie's Bay section (Twyning Mudstone Formation and Blue Anchor Formation) was assigned to the Norian. The Norian–Rhaetian boundary is now located at the top of the long normal SA3n (Fig. 6). Consequently, the upper half of the non-marine St. Audrie's Bay section (including the upper part of the Twyning Mudstone Formation and the entire Blue Anchor Formation) belongs to the Rhaetian.

4.4. Implication for a palynological characterization of the Norian–Rhaetian transition

To date, a biostratigraphic correlation of the Norian–Rhaetian transitional interval between the marine Tethys realm and the continental sequences (e.g. Germanic Triassic basin, Newark basin) is difficult also because of the paucity of independently dated palynological records (Kürschner and Herngreen, 2010). Unfortunately, palynological samples from the Steinbergkogel section proved to be barren (Krystyn et al., 2007a).

However, the presence of *Rhaetipollis germanicus* may be a valuable biostratigraphic marker Rhaetian stage as several palynological studies on alpine sections, such as Kendelbach and Weissloferbach (e.g. Morbey, 1975; Schuurman, 1979) have suggested. To date, *R. germanicus* has not yet been reported from the Arnstadt Formation (Steinmergelkeuper) which is traditionally assigned to the Norian (Nitsch, 2005). The exact entry level of *R. germanicus* is still unknown because no data exists from the Plattenkalk deposits underlying the Rhaetian Koessen or Zlambach beds. Furthermore, the lowest occurrence of *R. germanicus* in the Twyning Mudstone Fm of the St. Audrie's Bay section (Hounslow et al., 2004) appears in the reverse interval SA 3r. This chron has been previously assigned to the Norian and is now – based on the here presented magnetostratigraphic correlation – correlated to the Rhaetian. This is supported by the position of *R. germanicus* in the Postera beds (Schulz and Heunisch, 2005 and literature cited herein; Lund, 1977, 2003) of the German Keuper, which have been tentatively correlated to Sevatian 2 (Rhaetian) of the Alpine realm.

Based on our magnetostratigraphic correlation, the Norian–Rhaetian boundary closely correlates with the transition of the lower Passaic Heidlersburg to the Upper Balls Bluff–Upper Passaic palynofloral zone in the Passaic Formation of the Newark Basin (Cornet, 1977, 1993). *R. germanicus* has not been reported from North American Upper Triassic sequences. From the few palynological elements that are common in both, the North American and the European sections, *Granuloperculatipolis rudis* may be useful for palynostratigraphic correlation. This taxa is dominating the Rhaetian assemblages (Cirilli, 2010). It occurs in the middle and upper part of the Arnstardt Formation of the Germanic Triassic basin. In the Newark Basin, the FAD of *G. rudis* appears in the upper part of the Lower Passiac–Heidlersberg palynoforal zone (around E16n at ~212 Ma) and its LAD in the upper part of the Upper Balls Bluff–Upper Passaic palynofloral zone (around E21 at ~205 Ma) (Cornet, 1993; Fowell and Olsen, 1993). The position of the FAD of *G. rudis* in the Newark Basin, now closely correlated to the Norian–Rhaetian boundary. This would result in an early Rhaetian age for the upper part of the Arnstadt Formation, similar to the Twyning Mudstone Formation in Great Britain. However, the palynostratigraphic correlation based on *G. rudis* should be confirmed by further detailed studies from sections in both the marine and the continental realm.

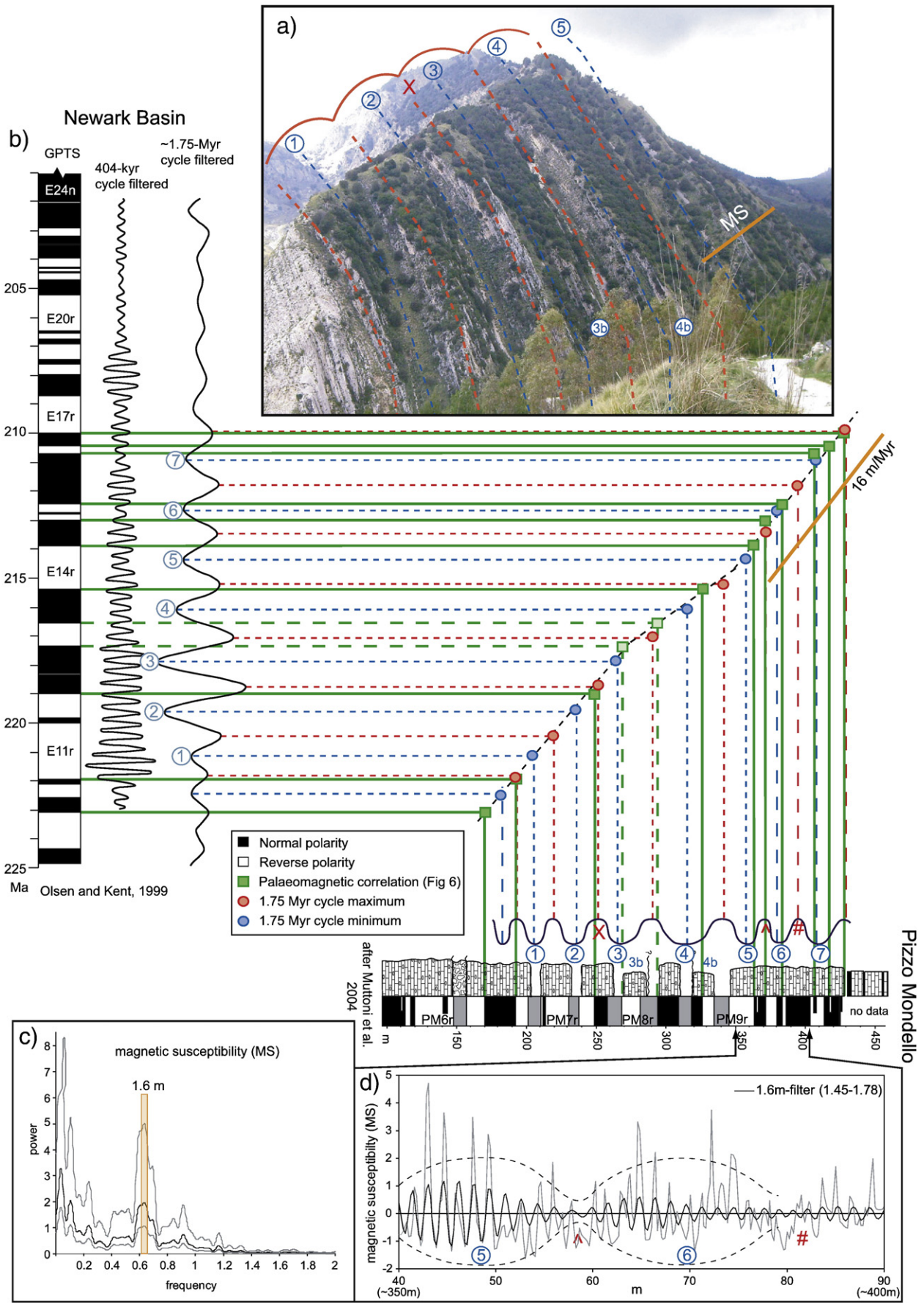
5. Cyclostratigraphic control on the age of the Norian–Rhaetian boundary — a future perspective

Regarding the long-stranding controversies of the Late Triassic time scale, it has become clear hat magnetostratigraphic and biostratigraphic correlations cannot unanimously solve the ongoing debate on the exact duration of the Rhaetian stage. The logical step forward is to integrate cyclostratigraphic information and focus on orbital tuning of Triassic marine sections. Based on cyclostratigraphy Kemp and Coe (2007) provided independent constraints on different palaeomagnetic correlations in a Triassic succession of the UK, while Ruhl et al. (2010) refined the duration of the early Jurassic stage previously calibrated by equal-duration of ammonite zones.

Here, we will explore the utilization of cyclostratigraphic constraints gained from the Norian–Rhaetian succession of the Pizzo Mondello section on Sicily, Italy for the chronology of the Late Triassic. In particular, the recognition of orbital parameters in the Norian succession will provide constraints on its duration and thus on the age and position of the Norian–Rhaetian boundary.

Tens-of-meters-scale cycles are evidenced in the lithologic succession of the Pizzo Mondello section on Sicily, which is characterized by distinct repetitions of indurated limestone intervals (Fig. 7a). Based on our field observation, we clearly recognize four

Fig. 7. a) The Late Triassic Pizzo Mondello section. Large scale cycles (1–7) are indicated, as well as the proposed ~1.75-Myr eccentricity maxima and minima. Symbols (red) and numbers refer to the same symbols used in b) and d). MS indicates the analyzed magnetic susceptibility in the section. b) Correlation of the Pizzo Mondello section (slightly modified after Muttoni et al., 2001, 2004) to the Newark section (Kent et al., 1995; Olsen and Kent, 1999) based on palaeomagnetic correlation (green; see Fig. 6) and correlation of ~1.75-Myr eccentricity maxima (red) and minima (blue). c) Power spectrum of time-series analysis of the magnetic susceptibility measured on the 350–400 m interval in the Pizzo Mondello section (our 40 to 90 m). Frequencies are given with 90% uncertainty intervals. d) Magnetic susceptibility (350–400 m) with the filtered 1.6 m cycle (bandwidth 1.45–1.8 m). The magnetic susceptibility data is de-trended and normalized before spectral analysis using Paillard et al. (1996). Symbols refer to the same symbols and numbers as used in a) and b).


a)
MS
3b
4b
Newark Basin
b)
GPTS
404-kyr cycle filtered
~1.75-Myr cycle filtered
E24n
E20r
E17r
E14r
E11r
205
210
215
220
225
Ma
Olsen and Kent, 1999
16 m/Myr
Normal polarity
Reverse polarity
Palaeomagnetic correlation (Fig 6)
1.75 Myr cycle maximum
1.75 Myr cycle minimum
after Muttoni et al. 2004
PM6r
PM7r
PM8r
PM9r
no data
Pizzo Mondello
m
c)
magnetic susceptibility (MS)
1.6 m
power
frequency
d)
1.6m-filter (1.45-1.78)
magnetic susceptibility (MS)
(~350m)
(~400m)

large-scale limestone–marl alternations between ~190 and ~350 m (as can be seen in the photograph in Fig. 7). Between 260 and 340 m we identify two additional but weakly pronounced indurated limestone intervals, which each form part of one larger-scale limestone–marl repetition (Fig. 7; layers 3b and 4b). Since the large-scale cyclicity cannot be directly observed in the upper 75 m part of the Pizzo Mondello section, we sampled this part with a high resolution (~0.25 m sample resolution) for magnetic susceptibility measurements. Measurements were performed at the Paleomagnetic laboratory at Utrecht University on a KLY-2 susceptometer (a weight-corrected-average from at least three measurements). For spectral analysis in the depth domain we select a 50 m interval spanning the interval between 350 and 400 m of the stratigraphic log of Muttoni et al. (2004), which excludes unexposed parts of the section. The analysis was carried out using the Analyseries program of Paillard et al. (1996). It revealed one significant (~90%) frequency peak corresponding to a cycle thickness of ~1.6 m (Fig. 7c). Furthermore, we recognize two weakly expressed maxima in the magnetic susceptibility record (5 and 6 in Fig. 7). In total, we thus recognize six large-scale (~25 to ~50 m) cycles: four are visible in the lithological log and are observed in the field and two are recognized in the magnetic susceptibility data.

Since biostratigraphic control indicates that the entire Norian (~17 Myr) is present at Pizzo Mondello, these large-scale cycles cannot be attributed to the ~100-kyr or ~400-kyr eccentricity cycle. The interval between ~180 and ~400 m contains six pronounced large-scale cycles and corresponds to ~11 Myr (Fig. 6).A 6:11 relation suggests a long-period eccentricity forcing, in particular of a ~1.8 Myr period. Olsen and Kent (1999) explored the presence of long-term climatic cycles in the Newark record by tuning the depth rank and colour records to the ~400-kyr eccentricity cycle. This resulted in highly significant periods of climatic precession modulations at periods of ~1.75 Myr. These cycles are directly calibrated to the Newark GPTS (Fig. 7b). It therefore seems likely, that the recognition of large-scale cycles in the Pizzo Mondello section represent the ~1.75-Myr eccentricity modulation as recorded in the Newark Basin.

To validate this assumption, we can correlate the Newark Basin GPTS to the Pizzo Mondello magnetostratigraphy and compare the filtered period at Newark with the sedimentary expression in the Pizzo Mondello section. The first order correlation is based on palaeomagnetic calibration (thicker green lines in Figs. 6 and 7b), which also yields sedimentation rates for the Pizzo Mondello section. It shows an increase from ~17 m/Myr in the interval between ~170 and ~270 m to ~29 m/Myr between ~270 and ~350 m, followed by a decrease to ~16 m/Myr in the uppermost ~60 m.

We then compare the position of indurated limestones at Pizzo Mondello (red lines in Fig. 7) to the filtered maxima of the ~1.75-Myr eccentricity time series of the Newark Basin stratigraphy (Kent and Olsen, 1999). This suggests a good agreement between the ~1.75-Myr eccentricity maxima and midpoints of indurated limestone intervals for the three large-scale cycles at ~190, ~220, and ~250 m, respectively in Pizzo Mondello. We further notice that the filtered 1.75-Myr eccentricity maxima at ~216.0 Ma and ~214.25 Ma correlate to the unexposed intervals between ~290 and ~340 m at Pizzo Mondello. This phase-relation-shift could be related to the ~1.75-Myr eccentricity cycle modulation of the ~404-kyr eccentricity cycle (ratio 1: ~4.3), which would eventually result in interference pattern and a shift of the ~404-kyr peaks relative to the ~1.75-Myr modulation with a ~404-kyr minimum occurring during ~1.75-Myr maximum. This transition could explain the double nature of the limestone interval of the large-scale cycle around ~290 and ~340 m. Using a sedimentation rate of ~16 m/Myr based on the age model, the filtered 1.6 m cycle in the interval between 350 and 400 m at Pizzo Mondello corresponds to ~100 kyr. This suggests an influence of the ~100-kyr eccentricity cycle on the magnetic susceptibility record (Fig. 7d). Counting the filtered (at least 12 to 14) and un-filtered (at least 16 to 18) cycles in the weakly expressed maxima (numbered 5 and 6 in Fig. 7) surprisingly suggests a long-term period close to ~1.75 Myr. We realize that our record is too short to recognize the influence of the long-period ~1.75-Myr eccentricity cycle. However, we tentatively correlate the two maxima in magnetic susceptibility to minima in the filtered ~1.75-Myr time series at ~213.4 Ma and ~211.6 Ma in Newark and minima in magnetic susceptibility to the filtered ~1.75-Myr eccentricity maxima at 212.4 Ma and 210.7 Ma, respectively (Fig. 7d).

On a further notice, the filtered 1.75-Myr eccentricity maxima with highest amplitudes in the interval of polarity zones E12 to E15 at Newark (~219–214 Ma) correlate surprisingly well to the interval with most pronounced large-scale cyclic alternations in Pizzo Mondello, supporting the magnetostratigraphic calibrations of Fig. 6.

We realize that our suggested orbital calibration of the Pizzo Mondello needs to be validated by further cyclostratigraphic analysis. However, the establishment of a continuous (geo-)chemical proxy record is hampered by un-exposed intervals in the succession. Nevertheless, we propose that the cyclostratigraphic constraint from the Pizzo Mondello section supports the position of the Norian–Rhaetian boundary at the top of E16n in Newark and supports a long duration of the Rhaetian.

6. Conclusions

We established a high-resolution magnetostratigraphy of the Steinbergkogel section in Austria, which contains the proposed Rhaetian GSSP and discuss the implications for the Geomagnetic Time Scale of the Late Triassic. The positions of the two conodont bio-event FO of *M. hernsteini* or FAD of *M. posthernsteini* – proposed to define the base of the Rhaetian – closely correlate with the top of the long normal polarity interval at the base of the section. The bio-magnetostratigraphic correlation to other marine Tethyan sections results in the calibration of this normal polarity zone to E16n of the Newark Basin GPTS. The Norian–Rhaetian boundary is now placed close to the top of E16n in the Newark GPTS and has an (astronomically intercalibrated) age of ~209.8 Ma. Additionally, the biostratigraphic subdivisions of the marine Norian and early Rhaetian are correlated to the Newark Basin GPTS resulting in an age of ~226.6 Ma for the Carnian–Norian boundary, ~216.4 Ma for the Lacian–Alaunian boundary and ~211.8 Ma for the Alaunian–Sevatian 1 boundary. Consequently, the duration of the Lacian is ~10 Myr, of the Alaunian is ~4.5 Myr, of the Sevatian 1 is ~2 Myr and of the Rhaetian is ~9 Myr. These results are supported by a cyclostratigraphic correlation between the Newark and Pizzo Mondello sequences, which for the first time, indicates the presence of long period eccentricity cycles (~1.75 Myr) in the Late Triassic of the Tethyan realm.

Finally, the results of our magneto-bio-cyclo-stratigraphic correlations are in substantial agreement with the recent results from the southern Alps of Muttoni et al. (2010) and indicate a long duration for the Rhaetian stage. This seems to contradict the hypothesis of a short duration of the Rhaetian stage (e.g. Cirilli et al., 2009; Gallet et al., 2007; Krystyn et al., 2007a,b).

We conclude that the next logical step forward in solving the debate on the exact duration and ages of Triassic stages and stage boundaries is to integrate magneto-bio- and cyclostratigraphic information with orbital calibration.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.epsl.2010.12.012.

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